CentricAI: Predicting Personal Finance Trends Using AI

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Introduction

Goal: Predict personal spending & investment trends to help users manage their finances more effectively.

Approach:

- Engineered 25+ behavioral & financial features from transactional data
- Built supervised learning models including Random Forests, Gradient Boosting, & LSTMs
- Managed the full AI lifecycle: data preprocessing → modeling → evaluation → deployment

Outcome:

- Achieved ~89% forecasting accuracy on validation datasets
- Identified key behavioral patterns driving spending & investment deviations

Visual Ideas:

- Small icon or mini diagram showing data → model → results workflow
- Bullet points to keep text scannable

Dataset

Source:

- Kaggle "Mock Personal Finance Data" & BankSim synthetic datasets

Preprocessing Steps:

- Cleaned missing or inconsistent transaction records
- Normalized numerical features for model input
- Created aggregated behavioral features (e.g., monthly spending trends, transaction frequency)

Dataset Stats:

FEATURE TYPE	COUNT/RANGE
Users	1,000
Transactions	100,000+
Time Span	12 Months
Features	25+ (financial & behavioral)

Problem Statement & Model Approach

Problem: Many individuals struggle to track, forecast, & manage personal finances, making budgeting & investment planning difficult.

Research Questions:

- How can AI models predict spending and investment patterns?
- Which behavioral and financial features drive budget deviations?

Context & Motivation:

- Synthetic datasets offered structure but lacked real-world variability, requiring careful preprocessing
- Imbalanced spending categories demanded normalization & targeted feature creation

AI Lifecycle & Modeling

Data Processing:

- Cleaned transaction data & engineered 25+ behavioral & financial features (frequency, volatility, income variability)

Models Used:

- Random Forest / Gradient Boosting: Strong on tabular, engineered features
- LSTM: Captured temporal spending trends

Strategy:

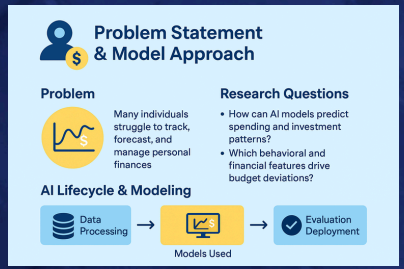
- Combined model types to learn both sequential patterns & behavior-driven signals for more stable forecasts

Evaluation & Deployment:

- Metrics used: Accuracy, RMSE, R^2, confusion matrix
- Modular pipeline built for scalability & deployment within CentricAI

Future Direction:

- Integrate real transaction data with privacy safeguards
- Extend forecasting to multi-year planning & anomaly detection



Results

Model Performance:

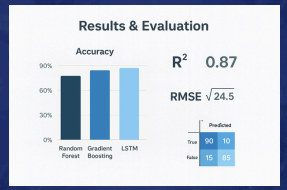
- LSTM achieved the highest overall forecasting accuracy (~89%), especially for temporal spending patterns.
- Gradient Boosting performed strongly on engineered behavioral features.
- Random Forest provided stable baseline performance with lower variance across runs.

Evaluation Metrics:

- R^2: 0.87 — strong correlation between predicted and actual spending
- RMSE: 24.5 — low average deviation in forecasting values
- Confusion Matrix: High true-positive and true-negative rates for budget deviation classification

Interpretation:

- Temporal models (LSTM) captured month-to-month spending cycles effectively
- Tree-based models excelled at identifying feature-driven patterns (e.g., spikes in transaction frequency)
- Overall, the ensemble of models provided robust & interpretable financial forecasts



Key Takeaways

- Behavioral and temporal patterns significantly improved forecasting accuracy, with LSTM models excelling at sequential trends & tree-based models capturing feature-driven insights
- High-quality feature engineering & a structured AI pipeline proved more impactful than model complexity alone, resulting in stable & interpretable financial predictions

Kaggle

LinkedIn

GitHub

Resume